

# Silver Wheel Replacement: Corrected Eval

## Short Answer

**For the corrected silver-wheel test, Reve production is the best visual result right now.**

Flux was useful as a strict masked inpaint control, but it is not the visual winner here: it darkens the silver rim, especially on C2 and N2. The first corrected comparison makes this obvious: Reve copies the target silver wheel more directly.

What changed from the previous report:

- Removed Qwen generation examples. The current `fal-ai/qwen-image-edit/inpaint` path does not receive the wheel reference image, so it is not relevant for “copy this exact wheel”.
- Removed the old wide model matrix from the main story. It mixed a flawed matte-black prompt with a silver reference image.
- Added the original wheels for C1, C2, and N2, so it is clear what was replaced.
- Added a clean Reve production vs Reve masked eval comparison.
- Left a clear slot for the next experiment: **OpenAI GPT Image**.

Current demo recommendation:

1. Show **Reve production** as the best result for tomorrow’s stakeholder demo.
2. Keep **Reve masked eval** as a controlled experiment, not the production path.
3. Keep **Flux** as the strict-mask baseline, not as the visual winner.
4. Add **OpenAI GPT Image** next using the same three-case layout.

## Inputs

All corrected runs use the same target wheel: the silver multi-spoke rim below. The left side shows the original wheels on each source car.

### What Was Replaced

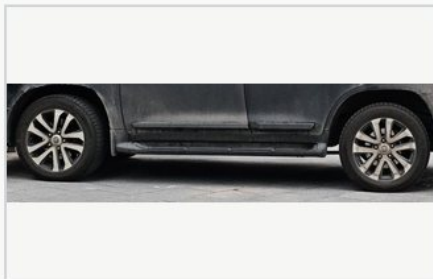
Each case starts with different original wheels. The target is one silver multi-spoke rim.

C1 - white car, daylight



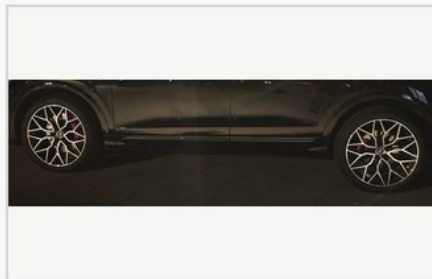
original wheels on the car

C2 - dark SUV, daylight



original wheels on the car

N2 - dark car, night



original wheels on the car

Target Wheel



silver reference

Corrected prompt principle:

Use the exact wheel design, color, finish, spoke pattern, center cap, and material from the reference image.

Preserve original scene lighting.

Do not alter car body, color, or background.

Important: the old prompt said **matte black** while the reference image was silver. That earlier visual report is useful only as a record of what happened, not as a fair silver-wheel comparison.

## Reve Production vs Reve Masked Eval

This is the key Reve comparison.

- **Reve production** receives car + wheel reference, no mask. This matches the current production request shape.
- **Reve masked eval** receives car + wheel reference + mask. This is an eval experiment, not the current production path.

### Reve: Production vs Masked Eval

Production gets car + wheel. Masked eval gets car + wheel + mask. Same corrected silver target.

Input



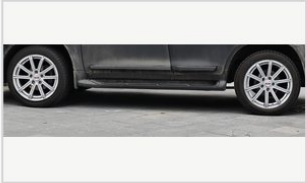
Reve production



Reve masked eval



Production zoom



# C1 - white car, daylight

One target wheel, three outputs. Reve production is the visual baseline; Flux is the strict-mask control.

Source Car



Original Wheels



what the model replaces

Target Wheel



silver reference

1. Reve production



car + wheel reference

2. Reve masked eval



car + wheel reference + mask

3. Flux masked control



mask + wheel reference

## Wheel Detail Zoom

Original



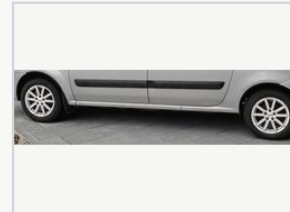
Target



Reve prod



Reve masked



Flux



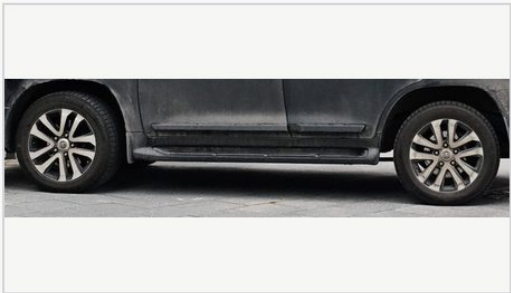
# C2 - dark SUV, daylight

One target wheel, three outputs. Reve production is the visual baseline; Flux is the strict-mask control.

Source Car



Original Wheels



what the model replaces

Target Wheel



silver reference

1. Reve production



car + wheel reference

2. Reve masked eval



car + wheel reference + mask

3. Flux masked control



mask + wheel reference

## Wheel Detail Zoom

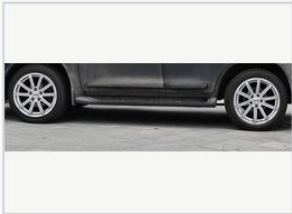
Original



Target



Reve prod



Reve masked



Flux



## N2 - dark car, night

One target wheel, three outputs. Reve production is the visual baseline; Flux is the strict-mask control.

Source Car



Original Wheels



what the model replaces

Target Wheel



silver reference

1. Reve production



car + wheel reference

2. Reve masked eval



car + wheel reference + mask

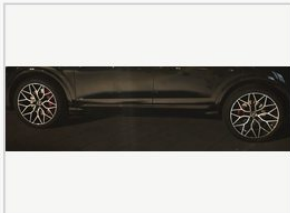
3. Flux masked control



mask + wheel reference

Wheel Detail Zoom

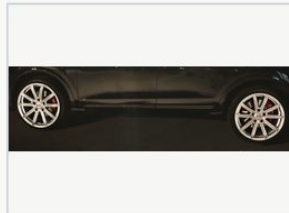
Original



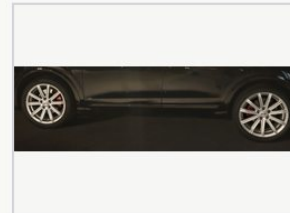
Target



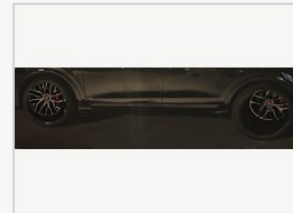
Reve prod



Reve masked



Flux



Model Status

Case takeaways:

- **C1: Reve production wins visually.** Reve masked eval is close. Flux keeps the edit localized but makes the target darker and less faithful.
- **C2: Reve production is the cleanest result.** Reve masked eval is usable but less clean. Flux loses the silver wheel identity.
- **N2: Reve production is still the best demo candidate.** It keeps the wheel silver and readable in low light. Flux collapses too far toward a dark rim.

| Path                              | Gets car | Gets wheel reference | Gets mask | Use in this report                       |
|-----------------------------------|----------|----------------------|-----------|------------------------------------------|
| Reve production                   | yes      | yes                  | no        | Main visual winner                       |
| Reve masked eval                  | yes      | yes                  | yes       | Side-by-side control                     |
| Flux Kontext inpaint              | yes      | yes                  | yes       | Strict-mask baseline                     |
| Qwen current fal inpaint          | yes      | no                   | yes       | Excluded                                 |
| OpenAI GPT Image via official API | planned  | planned              | planned   | Ready to run with OPENAI_API_KEY         |
| OpenAI GPT Image via AITUNNEL     | yes      | partial              | partial   | Blocked for honest masked reference test |

Why Qwen is out:

The current Qwen fal config receives only `image_url`, `mask_url`, and `prompt`. It does not receive the wheel reference image. That means it cannot honestly copy this exact silver rim.

Why Flux is not the winner:

Flux is better when the goal is “edit only inside this mask”. But the demo goal is “make the car look like it has this exact silver target wheel”. On the corrected examples, Reve production does that more clearly.

## Next: OpenAI GPT Image

Add OpenAI GPT Image as the next row in the same layout:

`source car + target wheel reference + alpha mask`

Do not change the cases, prompt intent, or visual layout. The next comparison should answer only one question:

Does OpenAI GPT Image beat Reve production on the same silver wheel target?

Suggested first run:

- C1, C2, N2
- same corrected target rim
- same adapted prompt
- medium quality first
- one high-quality retry only if the medium result is close

If GPT Image wins, it becomes the new demo candidate. If not, keep Reve production as the current best visual path.

AITUNNEL implementation status:

- The local runner now supports OpenAI-compatible AITUNNEL via `AITUNNEL_API_KEY` and `AITUNNEL_BASE_URL`.
- AITUNNEL `/v1/images/generations` works with `gpt-image-2`.
- AITUNNEL `/v1/images/edits` works for `image + mask` when the source image is sent as PNG.
- AITUNNEL `/v1/images/edits` also works for multiple `image[]` inputs without a mask.
- The required honest test shape, `image[] + mask`, currently fails through AITUNNEL before returning a JSON API error. Mixing `image` and `image[]` returns a normal 400 parameter error.

Conclusion: use the official OpenAI API for the controlled `car + wheel reference + alpha mask` GPT Image row, or ask AITUNNEL support to enable/fix `image[] + mask` passthrough for GPT Image edits.

## Run References

- Manifest: `tmp/fal-inpaint-eval/ivan-corrected-silver-3cases.jsonl`
- Reve production outputs: `tmp/reve-image-edit-eval/results-corrected-silver-production`
- Reve masked eval outputs: `tmp/reve-image-edit-eval/results-corrected-silver-masked`
- Flux masked baseline outputs: `tmp/fal-inpaint-eval/results-corrected-silver-flux`
- OpenAI/AITUNNEL runner: `scripts/openai_image_edit_eval.py`
- Presentation assets: `docs/assets/fal-mask-inpaint-eval/presentation`